

Building a Retail Sales Data Agent on Databricks

Student Name: Mandisa Nomonde Dumakude

Student Number: 2805001028

Date: 25 May 2026

1. Introduction & Objective

This project focused on building a Retail Sales Data Agent using Databricks. The aim was to upload a retail sales dataset, prepare the data, and configure an LLM powered agent capable of answering business-related questions using natural language queries. The project demonstrates how data analytics and AI tools can be combined to provide meaningful business insights from retail sales information.

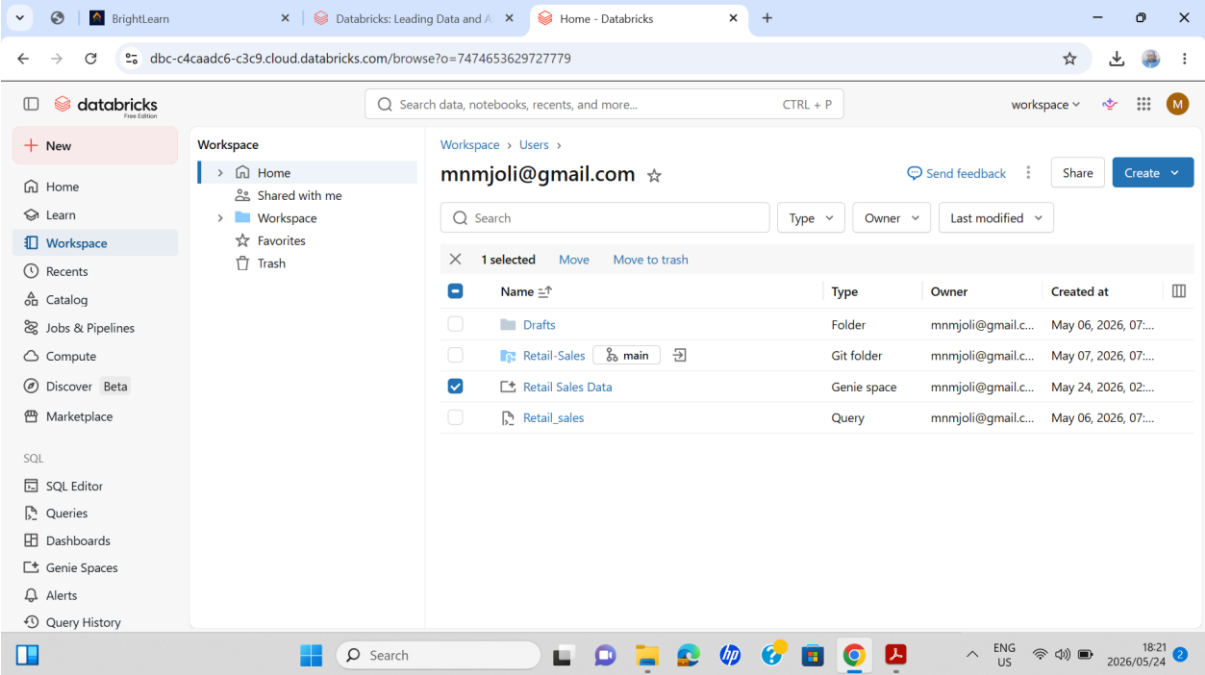
2. Dataset description

The dataset used in this project is called retail_sales_data. It contains transactional retail information such as customer ID, product categories, quantities purchased, prices, and total sales revenue.

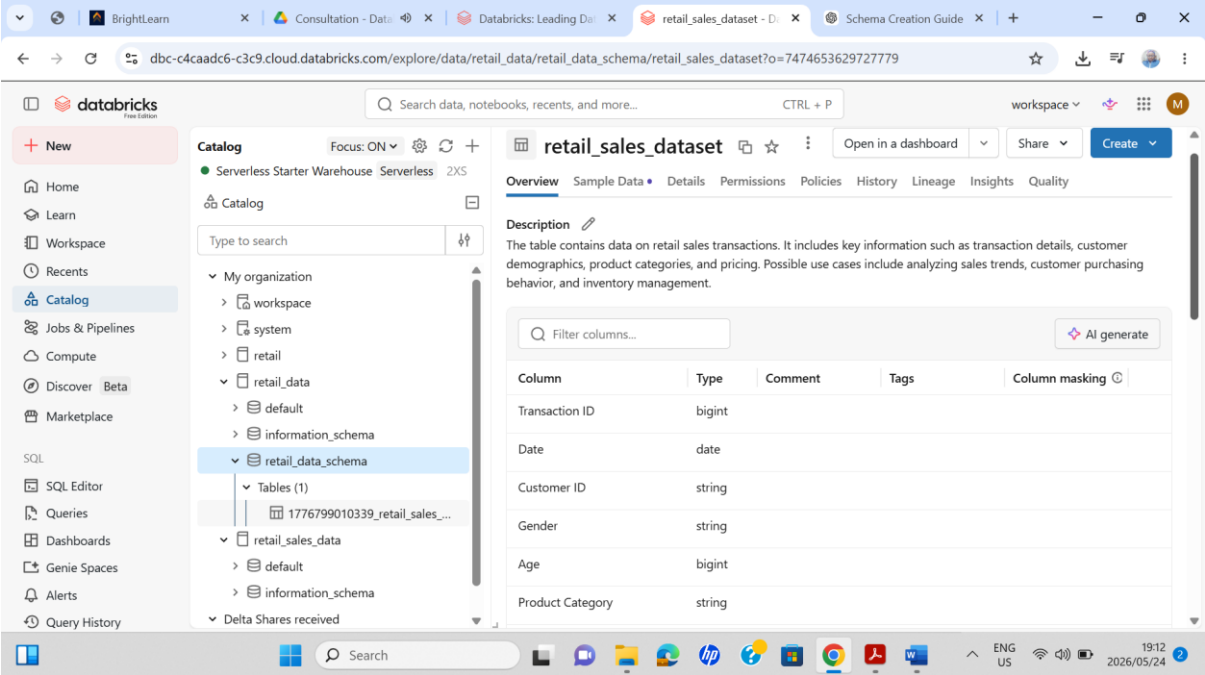
Column Name	Description
Transaction ID	Unique identifier for each transaction.
Date	Date on which the transaction occurred.
Customer ID	Unique identifier for each customer.
Gender	Customer gender (Male/Female).
Age	Customer age.
Product Category	Category of the purchased product.
Quantity	Number of products purchased.
Price per Unit	Price of a single product unit.
Total Amount	Total transaction value.

3. Screenshots

Upload Table:



Create schema:



Review data:

SQL Editor interface showing a query execution result for 'retail_sales_data'.

Query:

```
1 select *
2 from retail_sales_data;
```

Run all (1000) ✓ 3 minutes ago (<1s) retail_data: retail_data_schema

Transaction ID	Date	Customer ID	Gender	Age	Product Category
1	2023-11-24	CUST001	Male	34	Beauty
2	2023-02-27	CUST002	Female	26	Clothing
3	2023-01-13	CUST003	Male	50	Electronics
4	2023-05-21	CUST004	Male	37	Clothing
5	2023-05-06	CUST005	Male	30	Beauty
6	2023-04-25	CUST006	Female	45	Beauty
7	2023-03-13	CUST007	Male	46	Clothing
8	2023-02-22	CUST008	Male	30	Electronics

1,000 rows | 0.28s runtime | See performance (1) | Optimize | Refreshed 3 minutes ago

Agent setup:

Genie Spaces interface showing the 'Retail Sales Data' space setup.

Search data, notebooks, recents, and more... CTRL + P

workspace

Chat Monitor Benchmark Configure Share

Retail Sales Data

This space enables analysis of retail sales transactions to uncover sales trends, understand customer purchasing behavior, and support inventory management decisions.

Capabilities:

- Track key sales metrics such as total sales amount, transaction volume, and average...

Show more

Ask your question...

Agent Chat

Always review the accuracy of responses.

What is the distribution of customers' ages including minimum, maximum, average...

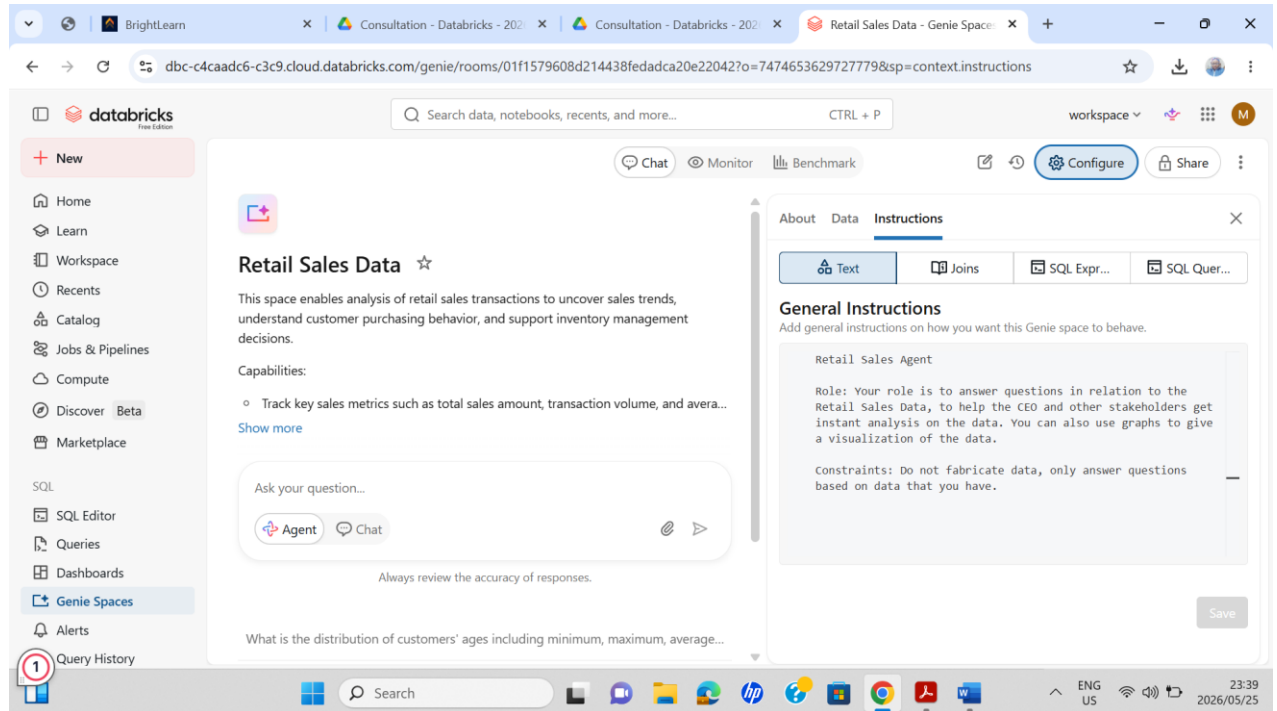
Data

Link existing or upload new data to start discovering insights.

Filter tables

Name	Type
retail_sales_data	Table
retail_data.retail_data_schema	Table

Agent instructions:



Full Agent instructions:

Retail Sales Agent

Role: Your role is to answer questions in relation to the Retail Sales Data, to help the CEO and other stakeholders get instant analysis on the data. You can also use graphs to give a visualization of the data.

Constraints: Do not fabricate data, only answer questions based on data that you have.

The 10 questions:

1. What is the total revenue generated across all products?

The image shows two screenshots of the Databricks interface. The top screenshot is the Genie Space, where a user asks "What is the total revenue generated?". The Genie Agent responds with "The total revenue generated is \$456,000.1.". The bottom screenshot is the SQL Editor, showing the same query executed: `SELECT SUM('Total Amount') AS total_revenue FROM retail_sales_data;`. The result table shows one row with a total revenue of 456000.

Genie Space Screenshot:

What is the total revenue generated ? Reviewable

Thinking complete

The total revenue generated is \$456,000.1.

Is this useful? ☐ ☐ Share Download PDF

Ask your question...

Agent Chat

Always review the accuracy of responses.

Instructions Panel:

General Instructions

Add general instructions on how you want this Genie space to behave.

Retail Sales Agent

Role: Your role is to answer questions in relation to the Retail Sales Data, to help the CEO and other stakeholders get instant analysis on the data. You can also use graphs to give a visualization of the data.

Constraints: Do not fabricate data, only answer questions based on data that you have.

SQL Editor Screenshot:

SQL Editor

Catalog Focus: ON

Type to search

My organization

- workspace
- system
- retail
- retail_data
 - default
 - information_schema
 - retail_data_schema
 - retail_sales_data2
 - default
 - information_schema
- Delta Shares received
 - samples

New SQL editor: ON

Run selected (1000) 2 minutes ago (<1s) retail_data.retail_data_schema

```
1 SELECT
2 SUM('Total Amount') AS total_revenue
3 FROM retail_sales_data;
```

Add parameter

Table	total_revenue
1	456000

1 row | 0.21s runtime See performance (1) Optimize Refreshed 2 minutes ago

2. Which product category generates the highest sales revenue?

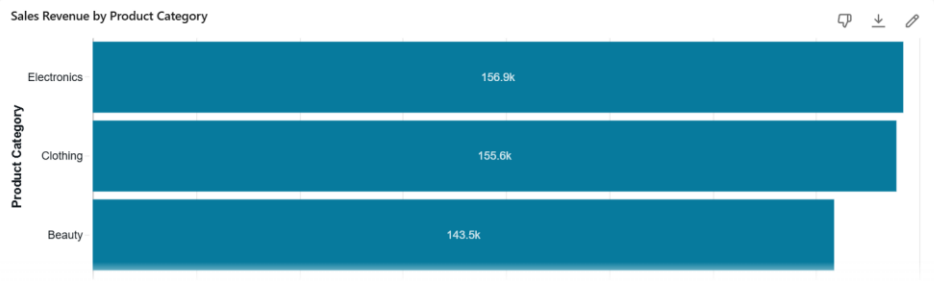
dbc-c4caadc6-c3c9.cloud.databricks.com/genie/rooms/01f1579608d214438fedadca20e22042/chats/01f158133e491ec3be46240b3fd65b04?o=7474653629727...

Search data, notebooks, recents, and more... CTRL + P workspace

2. Which product category generates the highest sales revenue? Reviewable Chat Monitor Benchmark Configure Share

Electronics generates the highest sales revenue with \$156,905 in total sales, slightly ahead of Clothing (\$155,580) and Beauty (\$143,515).¹

Sales Revenue by Product Category



Product Category	Total Revenue
Electronics	\$156.9k
Clothing	\$155.6k
Beauty	\$143.5k

Ask your question... Agent Chat

Always review the accuracy of responses.

dbc-c4caadc6-c3c9.cloud.databricks.com/editor/queries/3644839438285849?o=7474653629727779&contextId=sql-editor

SQL Editor Focus: ON Catalog

Type to search

- My organization
 - workspace
 - system
 - retail
 - retail_data
 - default
 - information_schema
 - retail_data_schema
 - retail_sales_data2
 - default
 - information_schema
 - Delta Shares received
 - samples

```
99 SELECT
100   "Product Category",
101   SUM("Total Amount") as total_revenue,
102   COUNT(DISTINCT "Transaction ID") as transaction_count,
103   SUM("Quantity") as total_quantity
104 FROM
105   "retail_data"."retail_data_schema"."retail_sales_data"
106 WHERE
107   "Product Category" IS NOT NULL
108 GROUP BY
109   "Product Category"
110 ORDER BY
111   total_revenue DESC
```

Run selected (1000) 10:09 PM (14s) retail_data:retail_data_schema

Add parameter

Product Category	total_revenue	transaction_count	total_quantity
Electronics	156905	342	849
Clothing	155580	351	894
Beauty	143515	312	849

3 rows | 13.91s runtime See performance (1) Optimize Refreshed 5 minutes ago

3. Which product category is performing the lowest?

The screenshot displays the Databricks Genie interface, which provides a conversational way to interact with data. The main panel shows the results of a query asking "Which product category is performing the lowest?".

Beauty is the Lowest Performing Product Category
Beauty is performing the lowest across all key metrics, generating **\$143,515 in total revenue** compared to Clothing (\$155,580) and Electronics (\$156,905).¹

Key Performance Metrics
Beauty underperforms in multiple dimensions:

- **Total Revenue:** \$143,515 (8.4% lower than Clothing, 9.3% lower than Electronics)
- **Transaction Count:** 307 transactions (12.5% fewer than Clothing's 351, 10.2% fewer than Electronics' 342)
- **Total Quantity Sold:** 771 units (13.8% lower than Clothing's 894, 9.2% lower than Electronics' 849)
- **Unique Customers:** 307 customers (the lowest among all categories)

A horizontal bar chart titled "Total Revenue by Product Category" shows Beauty with a red bar representing 143.5k.

Below the chart is a chat interface with an "Agent" button and a text input field "Ask your question...".

The bottom section of the image shows the SQL Editor interface. The "Catalog" pane on the left shows the database structure. The SQL query editor contains the following query:

```
SELECT
  'Product Category' as transaction_count,
  COUNT(DISTINCT 'Transaction ID') as transaction_count,
  SUM('Total Amount') as total_revenue,
  SUM('Quantity') as total_quantity_sold,
  AVG('Total Amount') as avg_transaction_value
FROM
  'retail_data`.`retail_data_schema`.`retail_sales_data`
WHERE
  'Product Category' IS NOT NULL
GROUP BY
  'Product Category'
```

The query results are displayed in a table with 3 rows and 4 columns: Product Category, transaction_count, total_revenue, and total_quantity_sold.

Product Category	transaction_count	total_revenue	total_quantity_sold
Beauty	307	143515	
Clothing	351	155580	

The table shows 3 rows and 4 columns. The first two rows are Beauty and Clothing. The third row is empty. The table is refreshed 2 minutes ago.

4. Which gender contributes more to the total sales?

The screenshot displays the Databricks Genie Space interface. The top section shows a chat window titled "Which gender contributes more to the" with a response stating: "Female Customers Contribute More to Total Sales. Female customers contribute slightly more to total sales, accounting for 51.06% of total revenue compared to males at 48.94%." Below this is a bar chart titled "Sales Contribution by Gender" showing Female with 232,840 and Male with 223,160. The bottom section shows the SQL Editor with a query and its results.

SQL Query:

```
SELECT
  Gender,
  SUM("Total Amount") as total_sales,
  COUNT("Transaction ID") as transaction_count,
  ROUND(
    SUM("Total Amount")
    / (
      SELECT SUM("Total Amount")
    )
    * 100.0
  ) as percentage_of_total
FROM
```

Query Results:

Gender	total_sales	transaction_count	percentage_of_total
Female	232840	510	51.06
Male	223160	490	48.94

5. Which product category has the highest sales for customers aged 20 and below?

The screenshot displays the Databricks workspace interface. The top section shows a Genie Space chat window with the question: "Which product category has the highest sales for customers aged 20 and below?". The chat response states: "Electronics has the highest sales for customers aged 20 and below, generating \$15,625 in total revenue from 23 transactions." Below the chat is a horizontal bar chart titled "Sales by Product Category (Customers Aged 20 and Below)". The chart shows three categories: Electronics (15.6k), Beauty (11.9k), and Clothing (7.3k).

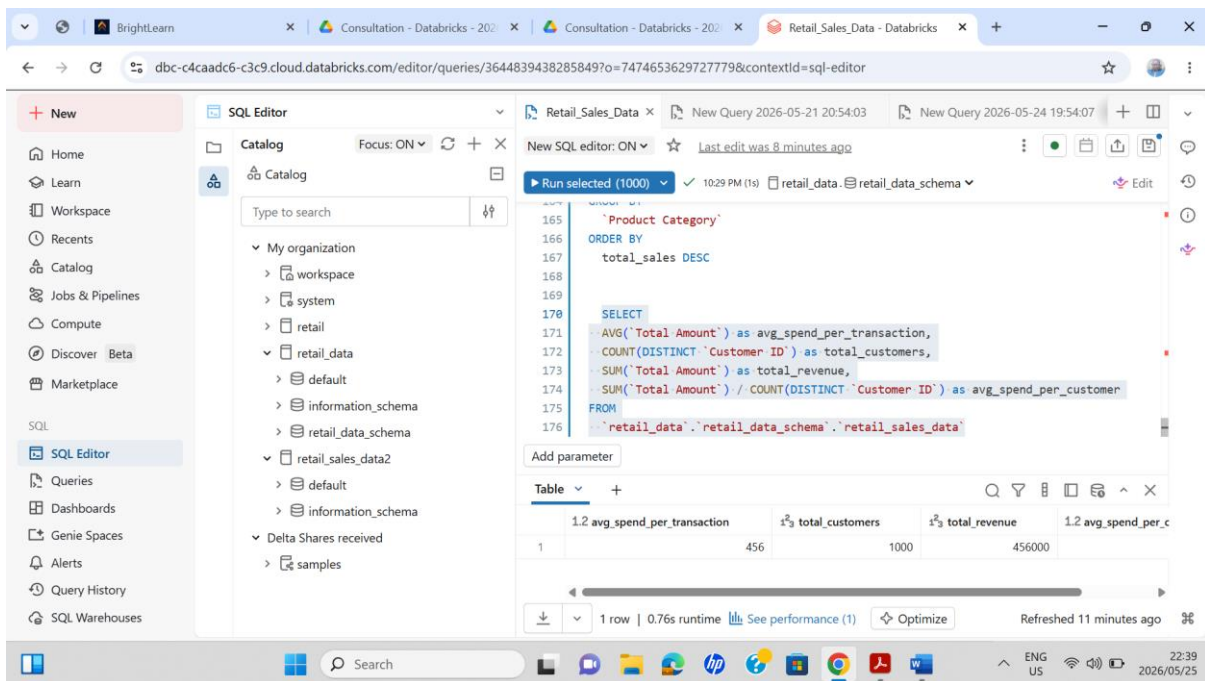
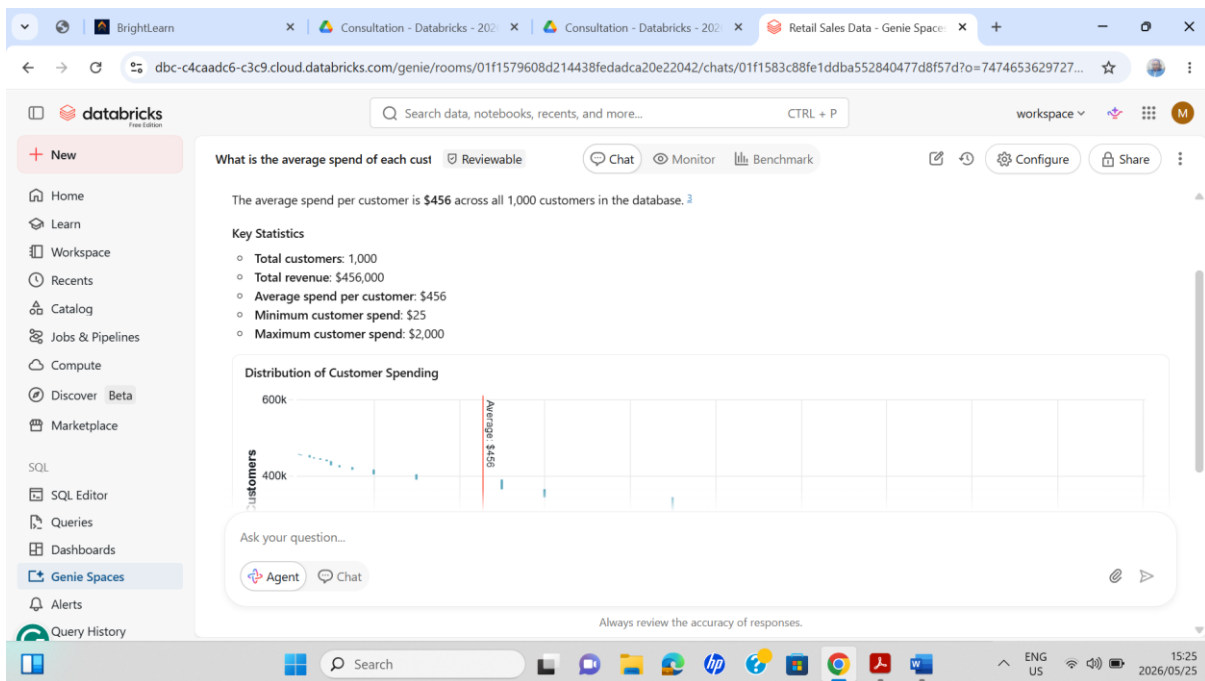
The bottom section shows the SQL Editor with a query that filters for customers aged 20 and below. The query is as follows:

```
SELECT
  'Product Category' as product_category,
  SUM('Total Amount') as total_sales,
  COUNT('Transaction ID') as transaction_count,
  SUM('Quantity') as total_quantity
FROM
  'retail_data`.`retail_data_schema`.`retail_sales_data`
WHERE
  'Age' <= 20
GROUP BY
  'Product Category'
ORDER BY
```

The query results are displayed in a table with 3 rows and 4 columns: Product Category, total_sales, transaction_count, and total_quantity.

Product Category	total_sales	transaction_count	total_quantity
Electronics	15625	23	60
Beauty	11850	24	60
Clothing	7255	16	38

6. What is the average spend of each customer?



7. What is the average spend for the top 10 customers?

The screenshot displays the Databricks Genie interface, which is used for interacting with data through natural language queries. The top section shows a chat window where a query has been processed. The query result indicates that the average spend for the top 10 customers is \$2,000. Below this, a detailed explanation states that all 10 top customers have exactly the same total spend of \$2,000, each having made just one transaction. This suggests these customers made single high-value purchases, likely of premium products. The interface also includes a search bar, a sidebar with navigation options, and a bottom section showing the SQL Editor with the corresponding SQL query.

What is the average spend for the top 10 customers?

Thinking complete

The average spend for the top 10 customers is **\$2,000**.

Interestingly, all 10 of the top customers have exactly the same total spend of \$2,000, with each having made just one transaction. This suggests these customers made single high-value purchases, likely of premium products.

Is this useful?

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Ask your question...

Agent Chat

Always review the accuracy of responses.

SQL Editor

Run selected (1000) 2 minutes ago (11s) retail_data.retail_data_schema

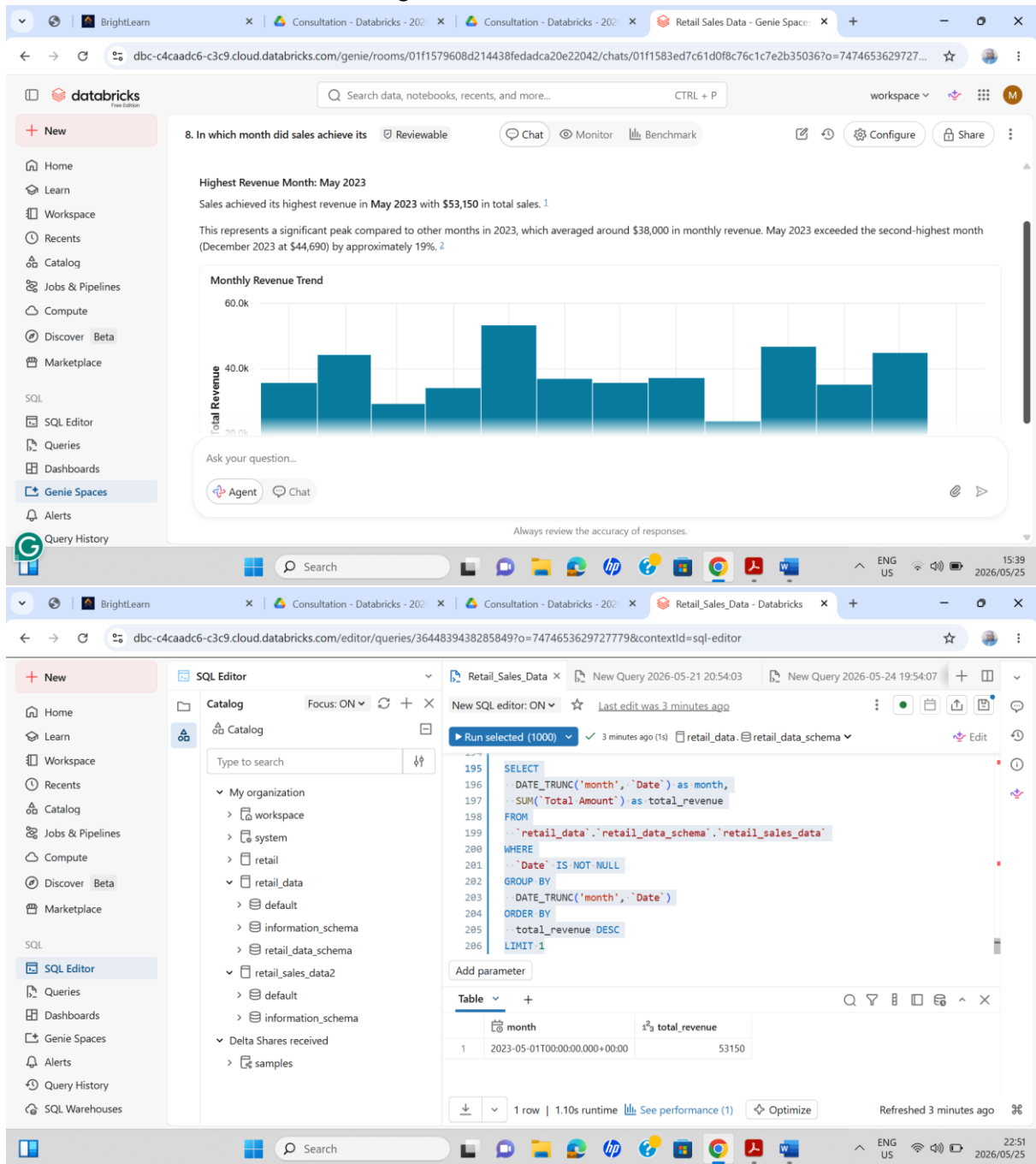
```
-- SELECT
-- Customer ID,
-- SUM('Total Amount') as total_spend,
-- AVG('Total Amount') as avg_transaction_amount,
-- COUNT(*) as transaction_count
FROM
  retail_data.`retail_data_schema`.`retail_sales_data`
WHERE
  Customer ID IS NOT NULL
GROUP BY
  Customer ID
ORDER BY
  total_spend DESC
```

Add parameter

Table	Customer ID	total_spend	avg_transaction_amount	transaction_count
1	CUST109	2000	2000	
2	CUST089	2000	2000	
3				

10 rows | 10.98s runtime | See performance (1) | Optimize | Refreshed 2 minutes ago

8. In which month did sales achieve its highest revenue?



The output is wrong, because the month is incorrectly displayed in the output.

9. Are there any noticeable purchasing patterns between men and women?

dbc-c4caadc6-c3c9.cloud.databricks.com/genie/rooms/01f1579608d214438fedadca20e22042/chats/01f1584393bf1992ba7bdfc202cae57d7o=7474653629727...

Search data, notebooks, recents, and more... CTRL + P workspace

Are there any noticeable purchasing pa Reviewable Chat Monitor Benchmark Configure Share

Gender Purchasing Patterns Analysis

Yes, there are distinct purchasing patterns between men and women, though overall spending behavior is remarkably similar. Women generate slightly higher total revenue (\$232,840 vs \$223,160) with more customers (510 vs 490), while average transaction values are nearly identical (\$456.55 for women vs \$455.43 for men). ²

Product Category Preferences

Women show a clear preference for Clothing, generating \$81,275 in revenue from 174 transactions, making it their top category. They distribute their purchases relatively evenly across all three categories (Clothing, Electronics, and Beauty). ¹

Men favor Electronics, generating \$80,170 in revenue from 172 transactions, making it their highest-revenue category. Notably, men make significantly fewer Beauty purchases (141 transactions) compared to women (166 transactions), representing an 18% gap. ¹

Total Revenue by Product Category and Gender

Ask your question... Agent Chat

Always review the accuracy of responses.

dbc-c4caadc6-c3c9.cloud.databricks.com/editor/queries/3644839438285849?o=7474653629727779&contextId=sql-editor

SQL Editor Focus: ON Catalog

Run selected (1000) 2 minutes ago (1s) retail_data: retail_data_schema

```
209 SELECT
210   'Gender',
211   COUNT(DISTINCT 'Transaction ID') as total_transactions,
212   COUNT(DISTINCT 'Customer ID') as unique_customers,
213   SUM('Total Amount') as total_revenue,
214   AVG('Total Amount') as avg_transaction_value,
215   SUM('Quantity') as total_quantity,
216   AVG('Quantity') as avg_quantity_per_transaction,
217   AVG('Price per Unit') as avg_price_per_unit
218 FROM
219   'retail_data`.`retail_data_schema`.`retail_sales_data`
220 WHERE
```

Add parameter

	Gender	total_transactions	unique_customers	total_revenue	1.2 avg
1	Male	490	490	223160	
2	Female	510	510	232840	

2 rows | 0.98s runtime See performance (1) Optimize Refreshed 3 minutes ago

10. Which age group spends the most in the store, 18-25, 26-35, 36-45, 46-60 and above 60?

The screenshot displays the Databricks Genie interface. The top section shows a chat window with the question: "Which age group spends the most in the store, 18-25, 26-35, 36-45, 46-60 and above 60?". The response states: "Age Group 46-60 Spends the Most. The 46-60 age group has the highest total spending at \$147,875, significantly outpacing all other age groups." Below the text is a horizontal bar chart titled "Total Spending by Age Group" showing the following data:

Age Group	Total Spending
46-60	147.9k
26-35	98.5k
36-45	91.9k
18-25	84.6k
Above 60	33.2k

The bottom section shows the SQL Editor with the following query:

```
SELECT
CASE
WHEN 'Age' BETWEEN 18 AND 25 THEN '18-25'
WHEN 'Age' BETWEEN 26 AND 35 THEN '26-35'
WHEN 'Age' BETWEEN 36 AND 45 THEN '36-45'
WHEN 'Age' BETWEEN 46 AND 60 THEN '46-60'
WHEN 'Age' > 60 THEN 'Above 60'
ELSE 'Other'
END AS age_group,
SUM('Total Amount') AS total_spending,
COUNT(DISTINCT 'Transaction ID') AS transaction_count,
COUNT(DISTINCT 'Customer ID') AS customer_count,
```

The query results are displayed in a table:

age_group	total_spending	transaction_count	customer_count
46-60	147875	331	331
26-35	98480	205	205

Validation of three answers:

1. The total revenue calculation
2. `SELECT`
3. `SUM('Total Amount') AS total_revenue`
4. `FROM retail_sales_data;`

2. Highest revenue product category

```
SELECT
  `Product Category`,
  SUM(`Total Amount`) as total_revenue,
  COUNT(DISTINCT `Transaction ID`) as transaction_count,
  SUM(`Quantity`) as total_quantity
FROM
  `retail_data`.`retail_data_schema`.`retail_sales_data`
WHERE
  `Product Category` IS NOT NULL
GROUP BY
  `Product Category`
ORDER BY
  total_revenue DESC
```

3. Highest spending age group

```
SELECT
  CASE
    WHEN `Age` BETWEEN 18 AND 25 THEN '18-25'
    WHEN `Age` BETWEEN 26 AND 35 THEN '26-35'
    WHEN `Age` BETWEEN 36 AND 45 THEN '36-45'
    WHEN `Age` BETWEEN 46 AND 60 THEN '46-60'
    WHEN `Age` > 60 THEN 'Above 60'
    ELSE 'Other'
  END AS age_group,
  SUM(`Total Amount`) AS total_spending,
  COUNT(DISTINCT `Transaction ID`) AS transaction_count,
  COUNT(DISTINCT `Customer ID`) AS customer_count,
  ROUND(AVG(`Total Amount`), 2) AS avg_transaction_amount
FROM
  `retail_data`.`retail_data_schema`.`retail_sales_data`
WHERE
  `Age` IS NOT NULL
GROUP BY
  age_group
ORDER BY
  total_spending DESC
```


Key insights

- Male and female purchase patterns was different
- Certain product categories generated higher sales than others
- Customer spending across age groups varied

Business recommendations

- Increase stock for high performing products
- Increase marketing on lowest performing products
- Focus more on frequent customers, offer discounts and promotions to increase sales even more by attracting new customers as well

Conclusion

The project successfully demonstrated how Databricks can be used to create an intelligent Retail Sales Data Agent. The agent was able to answer business questions, identify sales trends, and generate useful insights from the dataset. One challenge experienced was validating some outputs and ensuring date formats were correctly interpreted

GitHub Repository Link

<https://github.com/mndumakude/Building-a-Retail-Sales-Data-Agent-on-Databricks>